

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 8: Databases

Subtopic 8.3: Data Definition and Manipulation Languages (DDL & DML)

Concept 8.3.1: DDL Statements (CREATE, ALTER)

Total Questions: 13

Mark Schemes begin on Page 11

Question 1 | Source: 9618_w25_qp_12 (Q4.b)

- 4 A relational database, SHIPPING, stores data about the ships in a company and the containers that are carried on the ships.

The database has the following tables:

CONTAINER(ContainerID, Type, Weight, OwnerName, ShipID)

SHIP(ShipID, Type, Capacity, ShipName)

- (b) The table CONTAINER needs an additional field to store the data for the last inspection date.

Write a Structured Query Language (SQL) script to add **one** field to the table CONTAINER to store the date of last inspection of the container, for example 08/07/2019.

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Question 3 | Source: 9618_w24_qp_13 (Q4.c.i)

- 4 A company makes ice cream and sells it to shops.

The ice cream is made in batches: a large quantity of one type and flavour of ice cream that is then split into smaller quantities for sale.

The company’s owner has designed a relational database, ICECREAM, to store data about their ice cream and customers.

Some of the tables in the database are given. The database is not normalised.

BATCH(BatchID, Type, Flavour, Size, SellingPrice, EndDate)

CUSTOMER(CustomerID, CompanyName, EmailAddress, TelephoneNumber)

SALE(SaleID, BatchID, CustomerID, Quantity, Date)

- (c) The table definition for BATCH is repeated here:

BATCH(BatchID, Type, Flavour, Size, SellingPrice, EndDate)

Sample data for the table BATCH is given:

BatchID	Type	Flavour	Size	SellingPrice	EndDate
KIV12	Plain	Vanilla	1	2.20	12/12/2024
RIC14	Plain	Chocolate	0.5	1.80	12/12/2024
TYL1	Non-dairy	Lemon	0.5	2.10	01/02/2025
FYV2	Non-dairy	Vanilla	0.25	1.50	02/02/2025
BIV13	Plain	Vanilla	1	2.20	02/02/2024

(i) Write an SQL script to define the table `BATCH`.

Include constraints (restrictions) on the data that can be entered into each field where appropriate.

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Question 4 | Source: 9618_s24_qp_11 (Q6.c.i)

6 A company is developing a website that will allow users to create an account and then play a quiz every day. The data about the users and the quizzes are stored in a database.

A user must select a unique username and enter a valid email address to create an account. All users must be over the age of 16. A new quiz is given to the users every day. Each quiz is stored in its own text file.

The database stores the filename of each quiz and the date it can be played. The user gets a score for each quiz they complete, which is stored in the database. The scores are used to give each user a rating, for example Gold.

(c) The company has another database, `FARMING`, for a different game.

The database `FARMING` has a table named `EVENT` which is shown with some sample data.

PlayerID	EventID	Category	Points
000123	3	Build	100
000124	1	Grow	36
000123	4	Grow	22
000123	7	Create	158
000125	3	Grow	85
000125	4	Build	69

- (i) The database `FARMING` has a second table created named `PLAYER` that has the primary key `PlayerID`.

The field `PlayerID` in `EVENT` needs to be set up as a foreign key to link to `PlayerID` in `PLAYER`.

Write a Structured Query Language (SQL) script to change the table definition for `EVENT` to link the foreign key to `PLAYER`.

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Question 5 | Source: 9618_s24_qp_12 (Q4.b)

- 4 An assessment board wants to store the marks students achieved in exams in a database named `RECORDS`.

Part of the database design includes these two tables:

`EXAM(ExamID, Subject, Level, TotalMarks)`

`EXAM_QUESTION(ExamQuestionID, ExamID, QuestionNumber, Question, MaxMark)`

- (b) Sample data for the table `EXAM` is shown:

ExamID	Subject	Level	TotalMarks
00956124	Computer Science	2	75
00956125	Computer Science	3	120
00956126	Mathematics	2	100
00956127	Mathematics	3	150
00956128	Physics	2	70
00956129	Physics	3	80

Write a Structured Query Language (SQL) script to define the table `EXAM`.

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Question 6 | Source: 9618_s24_qp_12 (Q4.c)

- 4 An assessment board wants to store the marks students achieved in exams in a database named RECORDS.

Part of the database design includes these two tables:

EXAM(ExamID, Subject, Level, TotalMarks)

EXAM_QUESTION(ExamQuestionID, ExamID, QuestionNumber, Question, MaxMark)

- (c) The table EXAM_QUESTION has been created but the foreign key has not been linked.

Write an SQL script to update EXAM_QUESTION and link the foreign key to EXAM.

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Question 7 | Source: 9618_s24_qp_13 (Q4.b)

- 4 A theatre wants to use a database to store data about the shows that are scheduled, their customers and the seats that the customers have booked.

In the theatre:

- Each show can take place on multiple dates.
- Some dates can have more than one performance.
- There are multiple rows of seats.
- Each seat can be individually booked by its row letter and seat number, for example row E seat 2.

Part of the database design includes these tables:

SHOW(ShowID, Title, Duration)

SEAT(SeatID, RowLetter, SeatNumber)

PERFORMANCE(PerformanceID, ShowID, ShowDate, StartTime)

(b) Sample data for the table PERFORMANCE is shown:

PerformanceID	ShowID	ShowDate	StartTime
0001	MK12	5/5/2025	13:00
0002	MK12	5/5/2025	19:30
0003	MK12	6/5/2025	19:00
0004	OP3	7/5/2025	18:30
0005	OP3	8/5/2025	18:30
0006	OP3	9/5/2025	13:00

Write a Structured Query Language (SQL) script to define the table PERFORMANCE.

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Question 8 | Source: 9618_w23_qp_11 (Q3.c.i)

3 A shop manager has designed a relational database to store customer orders.

The database will have the following tables:

CUSTOMER(CustomerID, FirstName, LastName, Town)

SHOP_ORDER(OrderNo, CustomerID, OrderDate)

SUPPLIER(SupplierID, EmailAddress, TelephoneNumber)

ITEM(ItemNumber, SupplierID, Description, Price)

ORDER_ITEM(ItemNumber, OrderNo, Quantity)

(c) (i) Write a Structured Query Language (SQL) script to define the database called SHOP.

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Question 9 | Source: 9618_s23_qp_11 (Q2.b.iii)

2 An organisation uses a database to store data about the types of bird that people have seen.

- (b) The database, *Birds*, stores information about the types of bird and the people who have seen them.

Data about each bird seen is stored with its location and data about the person who saw the bird.

Database *Birds* has the following tables:

BIRD_TYPE(BirdID, Name, Size)

BIRD_SEEN(SeenID, BirdID, Date, Location, PersonID)

PERSON(PersonID, FirstName, LastName, EmailAddress)

- (iii) Part of the database table BIRD_TYPE is shown:

BirdID	Name	Size
0123	Blackbird	Medium
0035	Jay	Large
0004	Raven	Large
0085	Robin	Small

The database only supports these data types:

- character
- varchar
- Boolean
- integer
- real
- date
- time

Write a Structured Query Language (SQL) script to define the table *Bird_Type*.

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Question 10 | Source: 9618_w22_qp_11 (Q4.d)

- 4 A photographer creates a relational database to store data about photographs taken at birthday parties.

The database, PHOTOGRAPHS, stores details of the customer, the party, the photographs taken and the cameras used.

The photographer has several cameras that are used for taking the photographs at the parties.

Each camera has a specific lens type (for example, XY32Z) and lighting type (for example, F1672).

Data about each photograph is stored in the database including the party at which it was taken, the time it was taken and the camera used.

The database has these four tables:

CUSTOMER(CustomerID, FirstName, LastName, Telephone)

PARTY(PartyID, CustomerID, PartyDate, StartTime)

PHOTO_DATA(PhotoID, PartyID, TimeTaken, CameraID)

CAMERA_DATA(CameraID, LensType, LightingType)

- (d) Write an SQL script to include two new fields in CAMERA_DATA to store the number of photographs currently on the camera **and** the date the camera was last used.

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Question 11 | Source: 9618_w22_qp_13 (Q2.d)

- 2 The relational database ASTRONOMY is used to store data about telescopes, the companies that own the telescopes and the photographs taken by the telescopes.

The database has these three tables:

COMPANY(TelephoneNumber, CompanyID, CompanyName)

PHOTOGRAPH(PhotoID, TelescopeID, DateTaken, TimeTaken, Elevation)

TELESCOPE(TelescopeID, CompanyID, SerialNumber)

- (d) Write the SQL script to add **one** field to the table `PHOTOGRAPH` to store the resolution of the photograph, e.g. 1920 × 1068.

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Question 12 | Source: 9618_w21_qp_12 (Q6.c.ii)

- 6 A shop sells plants to customers. The shop manager has a relational database to keep track of the sales.

The database, `PLANTSALLES`, has the following structure:

`PLANT(PlantName, QuantityInStock, Cost)`

`CUSTOMER(CustomerID, FirstName, LastName, Address, Email)`

`PURCHASE(PurchaseID, CustomerID)`

`PURCHASE_ITEM(PurchaseID, PlantName, Quantity)`

- (c) The shop manager uses both Data Definition Language (DDL) and Data Manipulation Language (DML) statements to create and search the database.

- (ii) Write DDL statements to include a field in the table `PURCHASE` to store the date of the order.

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Question 13 | Source: 9618_s21_qp_11 (Q7.b.iii)

- 7 Bobby and Kim are discussing databases.

- (b) Bobby has a shop that sells products to customers. His database will store data about his customers, their payment details, orders and the products he sells. Customers will have login details to access their accounts. The database will update customers' payment and login details without keeping any historical records.

(iii) Bobby wants to name his database SHOPORDERS.

Write a Data Definition Language (DDL) statement to define a new database with the name SHOPORDERS.

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Question 14 | Source: 9618_s21_qp_12 (Q1.c.i)

- 1 Raj owns houses that other people rent from him. He has a database that stores details about the people who rent houses, and the houses they rent. The database, HOUSE_RENTALS, has the following structure:

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CUSTOMER(CustomerID, FirstName, LastName, DateOfBirth, Email)
HOUSE(HouseID, HouseNumber, Road, Town, Bedrooms, Bathrooms)
RENTAL(RentalID, CustomerID, HouseID, MonthlyCost, DepositPaid)
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(c) Example data from the table RENTAL are given:

RentalID	CustomerID	HouseID	MonthlyCost	DepositPaid
1	22	15B5L	1000.00	Yes
2	13	3F	687.00	No
3	1	12AB	550.00	Yes
4	3	37	444.50	Yes

- (i) Complete the following Data Definition Language (DDL) statement to define the table RENTAL.

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CREATE ..... (
    RentalID INTEGER NOT NULL,
    CustomerID INTEGER NOT NULL,
    HouseID ..... (5) NOT NULL,
    MonthlyCost ..... NOT NULL,
    DepositPaid BOOLEAN NOT NULL,
    ..... (RentalID)
);
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MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_12 (Q4.b)

4(b)	1 mark per bullet point, max 2 marks • ALTER TABLE statement • Including the additional field with a suitable field name and suitable field type Example answer ALTER TABLE CONTAINER ADD InspectionDate DATE;	2
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Mark Scheme for Question 2 | Source: 9618_w25_ms_13 (Q5.b)

5(b)	1 mark per bullet point, max 4 marks • Create table with opening and closing brackets and commas separating the attributes • Appropriate data types for StaffFirstName, StaffLastName and Department • Appropriate data types for StaffID and RemoteWorker • Primary key correctly defined Example Answer 1: CREATE TABLE STAFF(StaffID INTEGER, StaffFirstName VARCHAR, StaffLastName VARCHAR, Department CHAR, RemoteWorker BOOLEAN, PRIMARY KEY (StaffID)); Example Answer 2: CREATE TABLE STAFF(StaffID INTEGER NOT NULL PRIMARY KEY, StaffFirstName VARCHAR, StaffLastName VARCHAR, Department CHAR, RemoteWorker BOOLEAN);	4
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Mark Scheme for Question 3 | Source: 9618_w24_ms_13 (Q4.c.i)

4(c)(i)	1 mark for each bullet point • Create table BATCH with opening and closing brackets, all statements within brackets • BatchID, Type and Flavour as varchar or equivalent • ...with suitable constraint(s) • Size as decimal, SellingPrice as currency, EndDate as date (or equivalent) • Primary key identified as BatchID e.g. <pre>CREATE TABLE BATCH(BatchID VARCHAR(6) NOT NULL, Type VARCHAR (20) NOT NULL, Flavour VARCHAR (20) NOT NULL, Size FLOAT, SellingPrice CURRENCY, EndDate DATE, PRIMARY KEY(BatchID));</pre>	5
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Mark Scheme for Question 4 | Source: 9618_s24_ms_11 (Q6.c.i)

6(c)(i)	1 mark for each correct clause: • Alter table EVENT • Adding foreign key as PlayerID referencing correct table <pre>ALTER TABLE EVENT ADD FOREIGN KEY(PlayerID)REFERENCES PLAYER(PlayerID);</pre>	2
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Mark Scheme for Question 5 | Source: 9618_s24_ms_12 (Q4.b)

4(b)	1 mark each: • Creating table EXAM with opening and closing brackets • All fields with appropriate data types and commas at end of lines • ExamID as primary key Example: <pre>CREATE TABLE EXAM(ExamID varchar NOT NULL, Subject varchar, Level int, TotalMarks int, PRIMARY KEY(ExamID));</pre>	3
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Mark Scheme for Question 6 | Source: 9618_s24_ms_12 (Q4.c)

4(c)	1 mark each: - Altering table EXAM_QUESTION - Linking ExamID to ExamID in EXAM Example. <pre>ALTER TABLE EXAM_QUESTION ADD FOREIGN KEY (ExamID) REFERENCES EXAM(ExamID);</pre>	2
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Mark Scheme for Question 7 | Source: 9618_s24_ms_13 (Q4.b)

4(b)	1 mark each for: - Creating table PERFORMANCE with opening and closing brackets - Setting all four attributes with appropriate data types - Setting PerformanceID as primary key - Setting ShowID as foreign key referencing SHOW table Example: <pre>CREATE TABLE PERFORMANCE (PerformanceID varchar NOT NULL, ShowID varchar, ShowDate Date, StartTime Time, PRIMARY KEY (PerformanceID), FOREIGN KEY (ShowID) REFERENCES SHOW (ShowID));</pre>	4
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Mark Scheme for Question 8 | Source: 9618_w23_ms_11 (Q3.c.i)

3(c)(i)	1 mark for <pre>CREATE DATABASE SHOP;</pre>	1
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Mark Scheme for Question 9 | Source: 9618_s23_ms_11 (Q2.b.iii)

2(b)(iii)	1 mark each - CREATE TABLE start and end bracket - Bird ID as CHAR/VARCHAR - Name and size as VARCHAR/CHAR - Bird ID as primary key Example answer: <pre>CREATE TABLE BIRD_TYPE (BirdID CHAR(4) NOT NULL, Name VARCHAR(9), Size VARCHAR(6), PRIMARY KEY (BirdID));</pre>	4
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Mark Scheme for Question 10 | Source: 9618_w22_ms_11 (Q4.d)

4(d)	1 mark for each bullet point: - ALTER TABLE CAMERA_DATA - ADD NumberStored INTEGER , LastUsed DATE; <pre>ALTER TABLE CAMERA_DATA ADD NumberStored INTEGER, LastUsed DATE;</pre>	3
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Mark Scheme for Question 11 | Source: 9618_w22_ms_13 (Q2.d)

2(d)	1 mark for each bullet point: - ALTER TABLE PHOTOGRAPH - ADD Resolution TEXT; <pre>ALTER TABLE PHOTOGRAPH ADD Resolution TEXT / VARCHAR(11);</pre>	2
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Mark Scheme for Question 12 | Source: 9618_w21_ms_12 (Q6.c.ii)

6(c)(ii)	1 mark per bullet point - ALTER TABLE PURCHASE - ADD OrderDate - Suitable data type, e.g. DATE <pre>ALTER TABLE PURCHASE ADD OrderDate DATE;</pre>	3
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Mark Scheme for Question 13 | Source: 9618_s21_ms_11 (Q7.b.iii)

7(b)(iii)	1 mark <pre>CREATE DATABASE SHOPORDERS;</pre>	1
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Mark Scheme for Question 14 | Source: 9618_s21_ms_12 (Q1.c.i)

1(c)(i)	1 mark for each correctly completed line <pre>CREATE TABLE RENTAL (RentalID INTEGER NOT NULL, CustomerID INTEGER NOT NULL, HouseID VARCHAR (5) NOT NULL, MonthlyCost REAL/CURRENCY NOT NULL, DepositPaid BOOLEAN NOT NULL, PRIMARY KEY (RentalID));</pre>	4
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